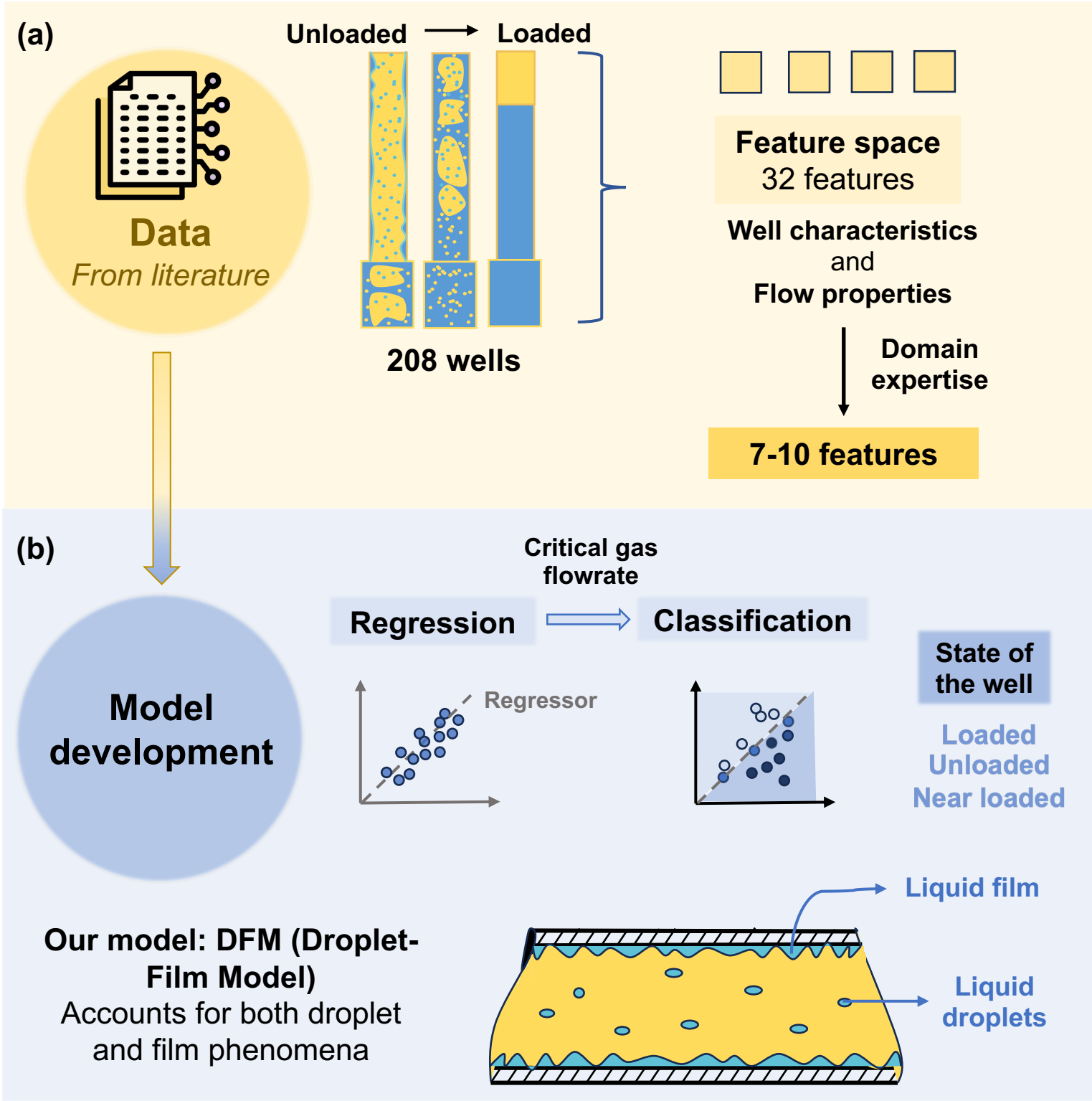
(a)
Data
From literature
Unloaded
Loaded
208 wells
Feature space
32 features
Well characteristics
and
Flow properties
Domain
expertise
7-10 features
(b)
Model
development
Critical gas
flowrate
Regression
Classification
Regressor
State of
the well
Loaded
Unloaded
Near loaded
Our model: DFM (Droplet-
Film Model)
Accounts for both droplet
and film phenomena
Liquid film
Liquid
droplets